ISRO-GeoNLI Report
Team - 46

Anonymous submission

Abstract

Satellite image analysis requires accurate semantic understanding and reliable spatial reasoning, under varying image resolutions and scales. Currently, Vision-Language Models (VLMs) are state-of-the-art architectures for combining vision and language modalities to achieve these tasks. However, existing VLMs suffer with problems such as hallucination, limited context window, fixed-resolution vision encoders, and weak vision-language alignment. To address these challenges, we develop a novel pipeline for Captioning, semantic, numeric and binary Visual Question Answering (VQA), and Grounding, on Remote-Sensing (RS) images. We utilize Qwen-based models which have native-resolution Vision-Transformers (ViTs). We also curate our own datasets, synthesize adversarial questions and perform counting-specific supervision. We also provide a thorough analysis of failure modes in existing models and demonstrate improvements in semantic robustness achieved through our scalable methodology.

Introduction

Satellite imagery has emerged as a critical data source for a broad spectrum of geospatial applications, including environmental monitoring, disaster response, urban planning, precision agriculture, defence, and large-scale infrastructure assessment. In contrast to conventional natural photographic images, satellite images span a wide range of spatial resolutions (on the order of 0.1-10 m/pixel or coarser), exhibit substantial geometric variability, and contain a very high density of semantically meaningful objects. Extracting actionable information from such data, therefore, demands models that jointly support high-level semantic understanding (e.g., attribute recognition, scene description) and fine-grained visual-spatial reasoning (e.g., counting, localization, orientation, and spatial relationship inference).

Recent advances in Vision-Language Models (VLMs) have significantly improved unified visual-textual reasoning capabilities. However, the majority of these models are pretrained primarily on natural image corpora and are thus poorly adapted to the specific characteristics and requirements of remote sensing imagery. From the perspective of standard VLM pretraining regimes, satellite scenes are fundamentally out-of-distribution: they consist of top-down nadir views with markedly different texture statistics, spatial organization, and illumination patterns. Moreover, these scenes typically contain extremely dense configurations of small objects, resulting in semantic and spatial complexity that far exceeds that of commonly used natural-image benchmarks.

Benchmarks such as VRSBench [20], which include oriented bounding box grounding tasks, further highlight the inadequacy of existing models in localizing objects with arbitrarily rotated orientations and dense clutter.

Mainstream VLMs - notably LLaVA [21], and GPT-4V [29] despite having impressive reasoning capabilities, exhibit major limitations when used on remote sensing imagery. First, they rely on fixed size Vision Transformers (ViTs) with input sizes of 224-336 making them incapable of processing variable aspect ratios. While methods like tiling and token pruning can be applied, they also result in adverse effects as reported in [4]. Second, their transformer-based encoders impose short context windows, restricting the number of visual tokens and forcing lossy compressions that eliminate fine-grained information. Third, prior research has shown that general-purpose VLMs frequently exhibit visual hallucinations, often predicting objects based on spurious priors rather even when they are absent [2].

To address limitations such as overfitting and weak generalization in fine-tuning existing models, we develop a unified Geo-Spatial Multimodal system capable of high-resolution captioning, semantic VQA (attribute, binary, and descriptive), numeric VQA (counting, measurement-like questions), and text-driven grounding. Our system builds upon the long-context capabilities of Qwen3-VL-8B-Instruct [43] equipped with native-resolution ViTs (Qwen-ViT 2.5/3.0) capable of handling variable aspect ratios without resizing distortion. This gives our system the ability to process very high resolution imagery.

In particular, we address these challenges with the following key contributions:

- We extensively benchmark models available for remote sensing based applications and propose a pipeline for semantic tasks like captioning and attribute based VQA, and vision based tasks like grounding and counting.
- We curate a downsampled large-scale multi-resolution dataset with synthetically generated VQA, with extensive captioning, adversarial and numeric questions, along with scale and resolution variance.



Figure 1: Comparison of segmentation approaches on satellite imagery. LISA (left) produces coarse region-level masks. GroundingDINO (center-left) detects objects but lacks spatial awareness. SAM3 (center-right) segments instances with labels but has a low context window and shows inconsistent precision. Our pipeline (right) achieves precise **instance-level segmentation with spatial grounding** correctly identifying positional relationships

- For grounding, we introduce a hybrid pipeline combining Qwen for text-conditioned object proposal generation, and Segment-Anything (SAM3) [5] for high-quality segmentation refinement and orientation extraction.
- We finetune Qwen3-VL-8B-Instruct and Ovis2.5 [26] for Geo-Spatial datasets and validate across two datasets with different image sources to verify generalization.

Our approach works entirely offline, using open-weight models, and is suitable for deployment in on-premise environments such as ISRO facilities.

Related Work

Vision Language Models

General-purpose vision-language models (VLMs) have rapidly advanced since CLIP introduced large-scale contrastive image-text pre-training [34]. To handle multiple resolutions, NaViT [8] was introduced. Recent multimodal LLMs such as LLaVA, LLaVA-1.5, LLaVA-Next, MiniGPT-4, Qwen-VL and Qwen2-VL have demonstrated strong performance on image captioning, open-ended VQA and instruction following ([21], [16], [41]). These models typically employ a ViT backbone with a projection layer or adapter to align visual features with a pretrained LLM. However, they exhibit well-documented failures including hallucination, weak fine-grained object recognition, and poor performance under distribution shifts. Long-context multimodal architectures have recently emerged to process larger inputs, including Qwen-VL-Chat, InternVL, and Kimi-VL with extended context windows ([41], [1] and [43]).

Mask RCNN, DeTr and GroundingDINO are different approaches to grounding from proposal-based localization to object localization without fixed categories ([9], [6], [23]). CountingDINO and DinoV3 are recent vision backbones which extract robust, high-quality dense features ([31], [36]). Segmentation models include SAM and variants ([13], [35], [5]). Models which introduce a unique segmentation token for masking include LISA and similar approaches ([15], [33], [32], [24]).

Remote Sensing Literature

A wide range of models for Remote Sensing applications include RSVQA, RSGPT, GeoLlaVa-8k, GeoChat, SamGeo, GeoGround and Falcon. ([25], [11], [39], [14], [48], [45]). For example, GeoLLaVA-8K uses token pruning to handle ultra-high-resolution images, while GeoChat finetunes Vicuna-7b, the language backbone of LLaVA-1.5 for remote sensing imagery. VRSBench, XLRS-Bench, DIOR, DOTA, iSAID, NWPU-Captions and FAIR-1M are among datasets that include remote sensing images various tasks like captioning, grounding and visual question answering. ([20], [40], [18], [42], [46], [7], [37]).

Model Selection & Implementation

A key insight in our work is that the nature of tasks possible could be broadly classified into those that are more reliant on the language capabilities of the VLM (e.g., like captioning and semantic question answering) and those that require strong visual understanding like grounding and counting, which guides our decisions regarding model selection, implementation and pipeline creation.

Vision Language Model Selection

Most common implementations of VLMs involve either cross attention based models like BLIP and Q-Former ([17], [12]) or architectures that utilize an adapter to map from the visual latent space to the text latent space, and then concatenate the vision and text tokens. Recently the latter has grown increasingly popular with a large majority of VLMs using such an approach ([38], [43], [26]). Despite its popularity across Geo-Spatial resoning models ([48], [14]) recent work on spurious correlations in multimodal LLMs [10] shows that VLMs like LLaVA can hallucinate objects purely from contextual cues, e.g., cars being strongly associated with roads even when they are not present. We validate this based on cosine similarity based heatmaps of text embeddings and visual tokens, as well as with LogitLens ([28]). Such analysis coupled with observed hallucination rates in generations led us to prefer newer models like Kimi-VL and Qwen-VL

with improved vision performance and native support for variable aspect ratios. This choice is further corroborated by recent research emphasizing importance of the language backbone for performance over simply vision models or visual tokens alone [19]. Finally, the increased context window (upto 128k with YARN) of these models is beneficial to handle high resolution images (a single $2k \times 2k$ contributes to over 16,000 tokens). We prefer Qwen3-8b-VL-Instruct over Kimi-VL due to its competitive performance at a smaller parameter count enabling finetuning.

It can be noted that typical adapters (like the two-layer MLP from LLaVA) may not lead to visual tokens being of the same distribution as text tokens. AlignVLM explicitly forces this by mapping the visual features into probability distributions over the LLM's existing pretrained vocabulary embeddings [27]. Ovis2.5, built on Qwen3 utilizes a similar architecture and offers a competitive choice. Due to the significant costs associated with pre-training, we avoid changing vision encoders for RS-specific encoders, or stronger vision foundational models like DINOv3 ([36]).

Grounding Model Selection

We initially experimented with GeoGround [48] on VRS-Bench, however we observed poor generalization to other datasets like XLRS. We hypothesize that this may be due to the inadequacy of the cross-entropy formulation of such a task in comparison to more typical weighted sum of L1 and generalized Intersection over Union used in Detection Transformers. Foundational detection models like MaskedRCNNs and Detection transformers struggle from closed vocabulary while GroundingDINO suffers due to poor localization. We therefore shifted our focus toward models with stronger segmentation priors, leading us to SAM3, which supports text and coordinate-based prompting and benefits from training on 4M phrases and 52M masks covering more than 50x the concepts found in existing benchmarks. SAM3's detector and tracker also leverage vision-language inputs via the extensively trained Perception Encoder (PE) backbone [3], making it a more suitable foundation for our downstream tasks.

To address the need for semantically and spatially aware grounding, we first adopted a cascaded architecture combining Qwen3-8b-VL-Instruct with SAM3. Initially, we utilized point-level localization capabilities of Qwen3 [43] to predict bounding boxes that were subsequently refined by SAM3 to yield pixel-level masks. These masks were then converted to oriented bounding boxes. However, this pipeline exhibited notable limitations: SAM3 is constrained to near-square inputs and supports resolutions only up to approximately 1008×1008, making it highly sensitive to aspect ratio and unsuitable for large high-resolution satellite imagery, and Qwen3 had significant failure modes in identifying the initial region of interest. Naive implementation of the inverted cascade, i.e., Qwen choosing from the candidate masks from SAM3 led to errors because of the challenging coupled task of spatial reasoning while identifying appropriate attributes in a long context setting.

Another observation was that tasks involving textual and visual information require attention to itself and to each other. By concatenating both, causal self attention mediates such interactions. LISA (Language Instructed Segmentation Assistant, [15] defines an additional <SEG> token to its vocabulary which is used to condition a SAM based MaskFormer. Similar architectures like UniPixel ([24], GeoPixel ([30]) and Chain of Visual Thought ([32]) utilize these representations to enhance visual reasoning. In particular Chain of Visual Thought trains segmentation heads on four models achieving enhanced representations. The effectiveness of such an approach is validated by its scores on VRSBench even without finetuning.

Methodology & Architecture

Frontend Architecture

The frontend provides an interactive interface where users can log in, view and resume previous conversations through the chat-history panel, or initiate new chats to keep topics separated. It supports both light and dark UI modes and includes an image-upload module where users can provide high-resolution satellite images and query the chatbot, which returns AI-generated responses based on the uploaded content.

Software Backend Architecture

The backend exposes REST endpoints for handling conversations and retrieving stored history. The `/chat` route processes incoming messages, stores them in MongoDB, and forwards the query to the AI model for inference. The `/chat-history` and `/get-history` routes manage retrieval of past sessions and message threads, enabling seamless continuation of older chats. MongoDB serves as the persistent layer for user details, chat sessions, and message logs.

API and Licensing

The system exposes a RESTful API built with FastAPI (v0.115+) serving on port 8080, providing two primary endpoints for remote sensing image analysis: `POST /query` and `POST /process`. The `/query` endpoint accepts natural language queries with automatic LLM-based task classification returning simplified `text, image` JSON responses, while `/process` accepts structured requests enabling explicit multi-task batch processing.

The system operates under MIT license for pipeline code, Apache 2.0 for Qwen3-VL model components, and Meta's SAM license for segmentation functionality. Connectivity requirements include: (1) outbound HTTPS access to HuggingFace Hub (`huggingface.co`) for initial model downloads ($\sim$32GB for Qwen3-VL-8B, $\sim$3-4GB for SAM3), (2) GPU with 36GB+ VRAM and CUDA 12.4 support for inference, (3) standard HTTP/1.1 client with JSON serialization capability and (4) optional base64 encoding support for image transmission.

Natural Language Inference pipeline

As mentioned earlier, we broadly approach the task in terms of solving semantic tasks and visual tasks. All VLM instances utilized is a finetuned version of Qwen3-8b-VL-

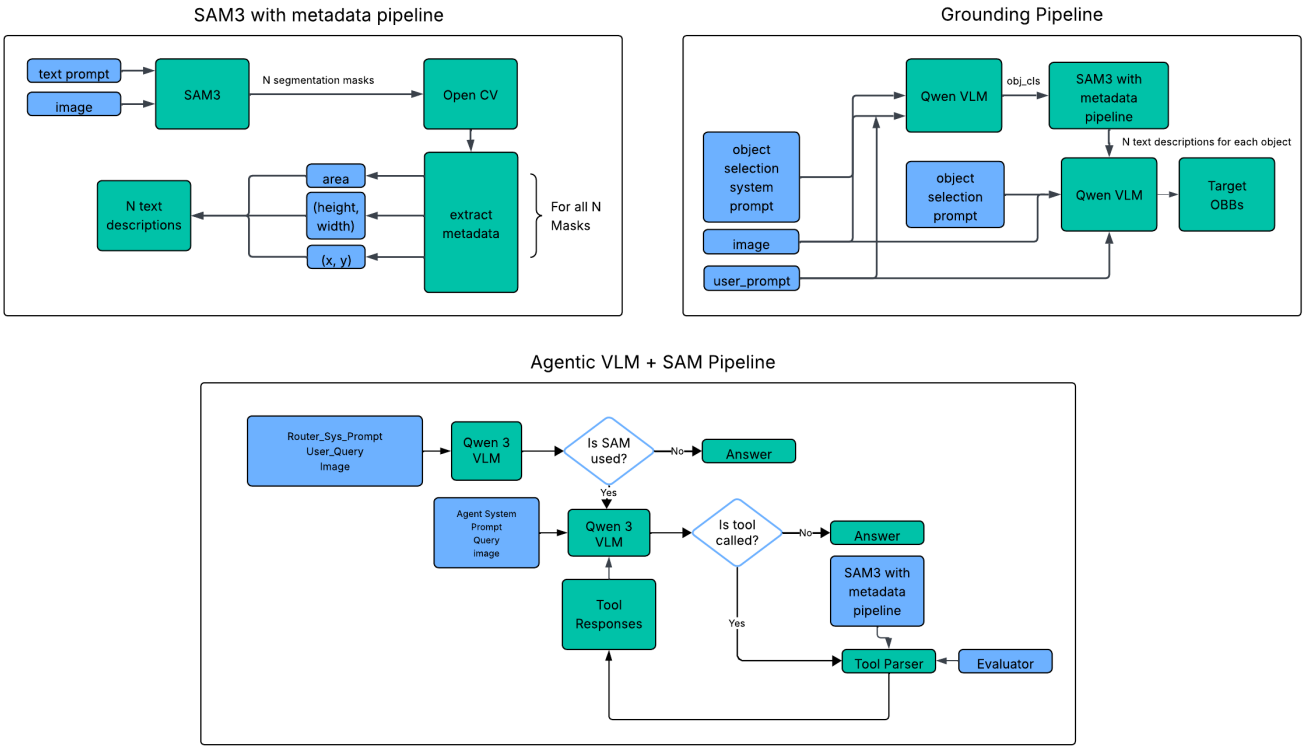


Figure 2: Grounding Pipeline and Agentic VLM + SAM Pipeline

Instruct trained on the train set of VRSBench and a doubly sampled curated XLRS Bench dataset. Details about dataset curation have been included in the following sections. Queries from the user are first routed to captioning, grounding or visual question answering tasks based on the intent inferred by the VLM (based on the provided query from the user alone). Captioning is handled by the same instance of the VLM with the image and the user query alone as context. The grounding task involves a multistage process which begins with the VLM being prompted to generate a highly descriptive short noun phrase which is then passed to SAM3. The segmentation masks from SAM3 are then programmatically parsed with OpenCV to extract metadata like position, orientation, compactness, width, height and area. These attributes are then parsed to generate a text description using heuristics, and the image is annotated with translucent segmentation masks to provide visual context. The VLM is then instructed to choose the appropriate object instance based on the query and image context provided. To improve the spatial inference capabilities of the VLM we annotate the image with gridlines and with set-of-mark prompting ([44]) by assigning each object an index and annotating the image appropriately. Alternatively we also provide an agentic pipeline where we completely bypass the need for the VLM to choose coordinates. We provide functions like `select_all`, `select_by_region`, `select_by_rank`, `select_top_n` and `select_by_threshold`. Sample functions and their descriptions are provided in the Appendix. Visual question answering questions are once again passed to a router that determines whether tool calling (including access to SAM3 to extract metadata) is necessary. This is done to utilize the native impressive performance of the finetuned VLM on semantic tasks while being able to offload visually challenging tasks to SAM3 and heuristics through an agentic tool calling pipeline. The tools available to this agent are: `detect_objects`, `get_object_info`, `measure_distance`, `calculator`. These descriptions are also provided in the appendix. Responses from the VLM are parsed and cleaned to meet the requirements posed by the response schema provided. Structured evaluation via the API does not utilize the initial router since the query type is has been provided.

Technical Details

The system implements several software engineering principles including modular architecture with clear separation of concerns across model interfaces (VLM, SAM3), task handlers (captioning, grounding, VQA), and API layers; singleton pattern for efficient model management to prevent redundant GPU memory allocation; dependency injection through constructor-based composition enabling flexible model swapping; and RESTful API design with FastAPI providing standardized endpoints and Pydantic schema validation. The codebase demonstrates defensive programming with comprehensive error handling and logging throughout

the pipeline.

Current limitations include synchronous request processing that blocks subsequent preventing concurrent execution and GPU parallelism, the singleton design with a single global pipeline instance preventing simultaneous multi-model serving and requiring server restarts for model changes, and missing request queuing that can allow uncontrolled concurrent requests to exceed the VRAM limit causing CUDA out-of-memory crashes under extreme load.

Experiments

Dataset Curation

Adapting VLMs to satellite imagery requires addressing the substantial domain gap between remote sensing data and natural-image pretraining corpora: satellite scenes are top-down, semantically dense, and contain large numbers of small objects, making conventional VLM assumptions unreliable. We also observed that prior remote-sensing VLMs such as GeoChat and GeoGround generalize poorly on VRS-Bench and XLRS-Bench. Further we notice that the distribution between the number of text and image tokens are highly skewed, with most datasets having about 20 times more image tokens than text tokens, which could lead to poor generalization by conditioning to the text alone.

To counter this, we constrain LoRA ranks to around (2-8, or $\sim$0.7-3.3M parameters) to match the scale of text supervision, by ensuring that the number of trainable parameters are of the same order of magnitude as the number of text tokens available. To maintain performance across resolutions, ground sampling distances and aspect ratios, we downsample ultra-high resolution images from XLRS. We further reduce spurious correlations by generating adversarial questions and fine-grained descriptive prompts using Gemini 2.5 Pro, ensuring that answers cannot be produced without examining the image. These adversarial samples-such as explicitly asking about the absence of cars on an empty road-force the model to rely on visual evidence, leading to more robust finetuning and significantly lower hallucination rates.

We perform various finetuning runs by training Qwen3-8b-VL-Instruct at different ranks, datasets and data mixtures. Our finetuning datasets include the train set of VRS-Bench ([20]), the curated train set of XLRS Bench ([40]) and RSVLM-QA ([49]). We provide statistics for the datasets and resolutions curated for in the tables that follow.

To improve the performance of the grounding pipeline, we attempted to finetune LISA for remote sensing tasks using a combination of remote sensing segmentation datasets. We curate datasets with pixelwise segmentation masks like GeoPix-Instruct[30], RRSIS-D[22], iSAID[47]. We apply tiling to the iSAID dataset as the images in that dataset are large and the masks are small.

Further, we fine train a separate model for SAR images derived from the SARLANG-1M dataset. It is a SAR image-text benchmark spanning 59 cities with hierarchical resolutions (0.1-25m), 1,696 object types, and 1,012 QA types across 7 applications. The curated dataset statistics are given in Tables 3 and 4.

GROUPED BY APPROXIMATE SIZE		
Size Category	**Count**	**Percentage**
$<$ 512px	9	0.06%
512-1K	765	4.85%
1K-2K	12143	76.93%
2K-4K	2022	12.81%
4K-8K	822	5.21%
8K-16K	24	0.15%

SUMMARY STATISTICS	
Statistic	**Value**
Total unique resolutions	1549
Most common resolution	1024x1024 (10178 images)
Second most common	4000x4000 (534 images)

Table 1: RSVLM-QA Curated Dataset Statistics

Aspect Ratio	**Resolution (W x H)**	**Count**
4:3	1024x768	250
3:4	768x1024	250
5:7	1000x1400	200
7:5	980x700	200
5:7	700x980	200
7:5	1400x1000	200
4:3	800x600	125
3:4	600x800	125
3:4	1200x1600	75
4:3	1600x1200	75
4:3	2048x1536	50
3:4	1536x2048	50
Square Resolutions (1:1)		
1:1	2048x2048	800
1:1	1024x1024	2000
1:1	768x768	800
1:1	512x512	400

Table 2: XLRS distribution of resolution and different aspect ratios

Split	**Images**	**Captions**	**VQA Pairs**
Train (66%)	8790	8716	56607
Val (33%)	4330	4291	27870
TOTAL	**13120**	**13007**	**84477**

Table 3: SAR VQA Captioning Dataset Split and Statistics (Avg VQA/Image = 6.44)

Image Resolution	Pixel Resolution (m/pixel)	Image Count
900 x 900	0.5	3311
512 x 512	0.5, 0.8, 1, 2	5222
1024 x 1024	0.15 - 0.5	4587

Table 4: SAR VQA Dataset Image Resolution Distribution

Captioning and VQA

We began with the hypothesis that pretraining an MLP-based connector could yield strong performance on semantics-heavy tasks such as image captioning and VQA. Motivated by this, we pretrained a two-layer MLP adapter bridging MoonViT's vision features and the Qwen3-4B language backbone. The adapter was trained for one epoch on PixelProse, Visual Genome, RSVQA, and NWPU Captions using four RTX 6000 Ada GPUs for approximately 15 hours.

However, human evaluation revealed several critical failure modes. The model frequently hallucinated visual details, produced repetitive or degenerate sentence structures, and often failed to answer queries correctly. A particularly severe issue was its inability to consistently generate the end-of-sequence $<\texttt{EOS}>$ token, resulting in unstable decoding and runaway text generation. These observations suggested that the MLP connector lacked sufficient inductive bias to reliably align MoonViT's visual representations with Qwen3's linguistic embedding space. In other words, simply learning a shallow mapping was insufficient for robust cross-modal alignment-leading us to consider that directly adapting the language model itself might be necessary.

This naturally led us to shift toward a finetuning-based alignment strategy, in which the LLM is partially updated to better accommodate the visual modality. Consequently, we performed two iterative finetuning experiments using the larger Qwen3-8B model.

- **Iteration 1:** We finetuned the last 18 layers of Qwen3-8B including attention blocks, MLP projections, and both simple and deepstack MLP mergers-using LoRA with rank 8 and $\alpha = 16$. Training was conducted for 3 epochs on the VRSBench training split, covering captioning, VQA, and grounding tasks.
- **Iteration 2:** Using the same parameterization and LoRA configuration, we performed a second finetuning stage focused exclusively on VQA. This stage used the XLRS, VRS, and RSVLM VQA datasets.
- **Iteration 3:** All the query and value matrices from all attention layers of Qwen3-8b were finetuned. Experiments included train set as only VRSBench[Train], only XLRS-Bench[Train], VRSBench[Train] + XLRSBench[Train] and VRSBench[Train] + 2 $\times$ XLRSBench[Train] across multiple ranks

Results for our approaches are available on table 8.

Thus, the final pipeline which we decided to go ahead with is given in figure 2. We also attempted finetuning Chain of Visual Thought by applying cross entropy loss only on the text tokens alongside model merging since the original training objective would have incurred signficiant costs and possibly weeks of compute time. However, due to the significantly different training approaches utilized, this did not lead to optimal convergence.

For grounding based approaches we extensively benchmark various traditional and state of the art approaches combined with tool calling. Other models benchmarked include SpaceSAM and LISAt.

Innovation & USP

Architectural Innovation

The proposed architecture is designed for computational and operational efficiency by decomposing multimodal inference into intent-routed, specialized pipelines. User queries are first routed based solely on inferred language intent, avoiding unnecessary visual processing and reducing early multimodal overhead. Long-form natural language queries are semantically compressed into concise noun phrases before invoking segmentation, improving grounding efficiency and segmentation recall. Expensive perceptual components such as object localization and spatial measurement are conditionally activated through an agentic tool-calling mechanism, ensuring that vision models are only executed when required. Spatial reasoning is formulated as symbolic object selection over structured metadata instead of direct coordinate prediction, improving robustness while reducing error propagation. Intermediate representations are reused across grounding and visual question answering tasks, preventing redundant computation.

Efficiency

To support training and evaluation on large-scale remote sensing datasets, we implement a scalable multi-task pipeline that performs just-in-time image loading and unified batching across captioning and VQA tasks. Instead of retaining decoded imagery in memory, the system stores only lightweight file references and loads images at batch construction time, substantially reducing memory overhead and preventing bottlenecks when processing thousands of high-resolution tiles. For supervised training on multiple tasks in one-shot, we implement a mutli-turn conversation setting which provides significant savings in compute and memory by avoiding expensive image loading multiple times.

For evaluation and response generation, the framework incorporates rank-aware dataset partitioning to enable efficient usage of multiple GPUs without cross-device synchronization.

We utilize Low Rank Adapters (LoRA) for parameter-efficient adaptation of vision-language models. We employ a selective LoRA strategy that injects adapters only into the query and value projections of self-attention layers in the text decoder, while excluding all visual encoder parameters. This targeted placement focuses learning capacity on the components most responsible for multimodal alignment, avoiding unnecessary modification of the computationally heavy visual tower. When combined with 4-bit quantization,

FlashAttention2, and gradient checkpointing, the approach enables efficient fine-tuning of multi-billion-parameter models on modest hardware while preserving their capacity to adapt to domain-specific captioning and VQA tasks. This selective parametrization is particularly effective for remote sensing, where visual features are stable but linguistic grounding requires task-specific refinement.

Results

Overfitting of Available Models in Captioning and VQA

Beyond architectural limitations, Geo-Spatial VLMs face unique failure modes not observed in natural-image tasks. We observed that the GeoGround model was overfit on the VRSBench dataset with 97% accuracy. However, it failed to generalize well on other datasets such as XLRS. Along with that, it has a poor context window and can only handle queries consisting a single sentence.

A key challenge, as highlighted by models such as GeoChat and GeoGround, is the severe token imbalance between vision and language streams. Remote sensing images may contain tens of thousands of visual tokens (e.g., 2k $\times$ 2k images at patch-size 16 have $\approx$ 16k tokens), while typical captions or QA labels contain only around 50 textual tokens. This imbalance results in poor multimodal alignment, text-overfitting, and weak generalization. Moreover, existing models lack explicit mechanisms for multi-scale reasoning, small-object discrimination, and orientation-aware grounding, all of which are central to satellite-image interpretation.

Grounding & Object Bounding

We conducted multiple experiments and inferenced multiple models on the VRSBench dataset. The results of the same are given in Table 5.

In the Qwen→SAM pipeline, most failures arise from Qwen predicting incorrect or overly large horizontal bounding boxes, which either misdirect SAM or reduce its segmentation accuracy. SAM's performance further degrades when the cropped regions are non-square, as its text-prompted segmentation is trained on 1008x1008 square inputs. Even when the bounding box is correct, SAM occasionally fails to segment the intended object within the provided region. In the SAM→Qwen pipeline, errors primarily occur when Qwen assigns the wrong target class to the objects segmented by SAM or selects a class that SAM cannot segment. SAM itself also struggles with non-square inputs for text-based segmentation. Additionally, when the annotated image is densely populated with objects, Qwen often misidentifies or incorrectly selects among the segmented regions, leading to the majority of failure cases.

Figure 1 gives a sample indication of our final pipeline over other approaches.

Captioning & VQA

The results of the same are given in Table 8. Our finetuned model performs significantly better after finetuning on VRS-Bench, further this performance translates well to XLRS Bench despite both of the datasets having different sources of images, density of objects and questions. The added manipulations in terms of scale and aspect ratio variation, adversarial and numeric questions make XLRS a particularly challenging benchmark. Generalization of adapters as low as rank 2 to a significantly harder task enables improved performance.

The agentic pipeline, tool calling and prompt specification brings improved performance to the final pipeline over any of their individual scores.

Mechanistic Interpretability Results

We analyzed the internal attention patterns of Qwen3-VL-4B-Instruct across 36 layers and 32 heads per layer. Our energy ratio analysis revealed that specific heads are highly specialized for spatial localization. As shown in Table 7, **Layer 20, Head 15** exhibited the strongest correlation with ground-truth objects, achieving an energy ratio score of 0.2697.

Qualitative analysis using the Logit Lens (Figure 3) confirmed that semantic concepts related to target objects (e.g., "stadium," "airport") begin to materialize in the vocabulary projection around Layer 15, preceding the peak spatial localization in Layer 20. This suggests a "recognize-then-localize" internal mechanism.

Acknowledgments

We gratefully acknowledge the ISRO Space Applications Centre (SAC) team for formulating the problem statement and guiding the direction of this work. We also thank the Organising Team at IIT Patna for their excellent coordination and consistent support throughout the project.

Appendix

Details of tools

The Visual Question Answering Agent has access to:

- detect_objects — Identifies all instances of a specified object class in the image and records their geometric and confidence attributes.
- get_object_info — Sorts detected objects by a chosen attribute, selects one by rank, and returns the requested property.
- measure_distance — Computes the centroid-to-centroid distance in meters between any two detected objects.
- calculate — Evaluates mathematical expressions for deriving aggregated or post-processed quantitative measures.

And grounding Agent has access to:

- select_all — Returns all detected masks, used for queries referring to all instances of an object class.
- select_by_region — Filters masks by a specified spatial region of the image.
- select_by_rank — Selects a single mask based on its rank when ordered by a chosen attribute.
- select_top_n — Retrieves the top N masks sorted by a specified attribute and order.

Model	IoU@0.3 (%)	IoU@0.5 (%)	IoU@0.7 (%)
SamGeo	36.36	20.20	11.11
LISA	18.42	14.29	10.12
UniPixel	38.67	25.52	7.62
Qwen 2.5VL 3b	62.07	38.45	18.90
Qwen 2.5VL 3b + SAM3	38.17	21.76	5.98
Qwen3 VL 8b Instruct + SAM	41.40	25.11	10.08
Method A	59.23	47.43	23.83
Method B	51.09	41.27	21.07
Method C	60.40	50.10	26.61
Method D	56.94	46.75	25.05
Method E	**59.93**	**49.81**	**28.06**

Table 5: IoU scores at thresholds 0.3, 0.5, and 0.7 for various grounding models. First 2 models evaluated on HBB subset of VRSBench. Post that all models evaluated on 500 images of VRSBench val OBB subset.
Method A - Qwen3-VL-8B-Instruct(prompt)+ SAM 3+ Qwen3 chooses best box (OBBs).
Method B - Qwen3-VL-8B-Instruct(prompt)+ SAM 3 + Qwen3 chooses best box (HBBs).
Method C - Qwen3-VL-8B-Instruct(prompt)+ SAM 3 - Annotated image is fed to Qwen3 with OBB metadata(image has a black border around it) + Qwen3 chooses best box (OBBs).
Method E - Qwen3-VL-8B-Instruct(prompt)+ SAM 3 - Annotated image is fed to Qwen3 with OBB metadata(image has grid lines) + Qwen3 chooses best box (OBBs).

Model	Captioning	Binary	Numeric	Semantic
Kimi	0.9290	0.6090	**0.5175**	0.8899
Geochat	**0.9344**	0.6192	0.4566	0.8571
Qwen 3 VL 8B (Finetuned)†	0.9200	0.7975	0.4757	0.9013
qwen-2-7B (4/8/2e-4/512)	0.8267	0.8415	0.3874	0.5463
qwen-3-VL-8B (2/4/2e-4/512)	0.8207	0.8411	0.5067	0.8404
qwen-3-VL-8B (4/8/2e-4/512)	0.8240	0.8483	0.5102	0.9127
qwen-3-VL-8B (8/16/2e-4/512)	0.8334	0.8544	0.5154	0.9362
Ovis-2.5-9B (Finetuned) ‡	0.8454	**0.9109**	0.5963	**0.9418**

Table 6: Comparison of Captioning, Binary, Numeric, and Semantic scores across various models on VRSBench.
Kimi evaluated on VRS Bench (1000 images). Geochat evaluated on VRS Bench (70 images).
† Qwen 3 VL 8B (Finetuned) configuration: LoRA rank=8, 18-36 Layers (K,Q,V) & Deepstackmerger + merger layers; evaluated on VRSBench Val (Counting VQA).
‡ Ovis-2.5-9B Finetuned on VQA and Captioning of VRSBench, XLRSBench, RSVLM for 3 epochs with LoRA Rank 8, all attention and mlp layers [23M Parameters] Qwen-series models evaluated using BERT-BLEU4 for Captioning and ISRO metrics for VQA tasks.

- select_by_threshold — Selects masks that satisfy an attribute-based threshold condition.

Rank	Layer	Head	Energy Ratio Score
1	20	15	0.2697
2	8	4	0.2639
3	13	10	0.2534
4	18	15	0.2502
5	9	18	0.2465

Table 7: Top 5 Attention Heads in Qwen3-VL correlated with Object Localization. Layers 18-21 show a consistent cluster of ”localization heads.”

Table 8: Experiment Run Results and Hyperparameters

Note: Bold indicates the highest score; Underline indicates the second highest score per column.

ID	Model	Rank	Alpha	Dataset	Tasks	VRSBench	XLRS-Bench
				Group A			
1	Baseline	-	-	-	Captioning, VQA	Caption: 0.68 Binary: 0.73 Numeric: 0.025 Semantic: 0.011	Caption: 0.79 Binary: **0.86** Numeric: 0.06 Semantic: 0
				Group B			
2	QV	8	16	VRSBench (Train)	Captioning, VQA	Caption: **0.83** Binary: **0.85** Numeric: 0.51 Semantic: **0.93**	Caption: 0.74 Binary: 0.70 Numeric: 0.23 Semantic: **0.87**
3	QV	16	32	VRSBench (Train)	Captioning, VQA	Caption: 0.81 Binary: <u>0.84</u> Numeric: **0.52** Semantic: **0.93**	Caption: **0.81** Binary: 0.75 Numeric: <u>0.27</u> Semantic: <u>0.78</u>
				Group C			
4	QV (epochs = 3)	8	16	VRS+XLRS+RSLVLM	Captioning, VQA	Caption: 0.67 Binary: 0.72 Numeric: 0.025 Semantic: 0.0172	Caption: 0.79 Binary: **0.86** Numeric: 0.0625 Semantic: 0
5	QV (epochs = 5)	8	16	VRS+XLRS+RSLVLM	Captioning, VQA	Caption: 0.66 Binary: 0.77 Numeric: 0.09 Semantic: 0.14	Caption: <u>0.8</u> Binary: **0.86** Numeric: 0.0896 Semantic: 0.0524
6	QV	8	16	VRS+XLRS x 2	Captioning, VQA	Caption: 0.80 Binary: 0.83 Numeric: 0.51 Semantic: **0.93**	Caption: **0.81** Binary: <u>0.78</u> Numeric: **0.29** Semantic: 0.72

Table 9: Experiment Run Results and Hyperparameters. All results are provided for `Qwen3-8b-VL-Instruct`. Notice the significant improvement in scores on training on VRSBench which generalizes to XLRS Bench despite XLRS having being more challenging due to varying resolutions, aspect ratios, number of objects and synthetically generated adversarial VQA pairs.

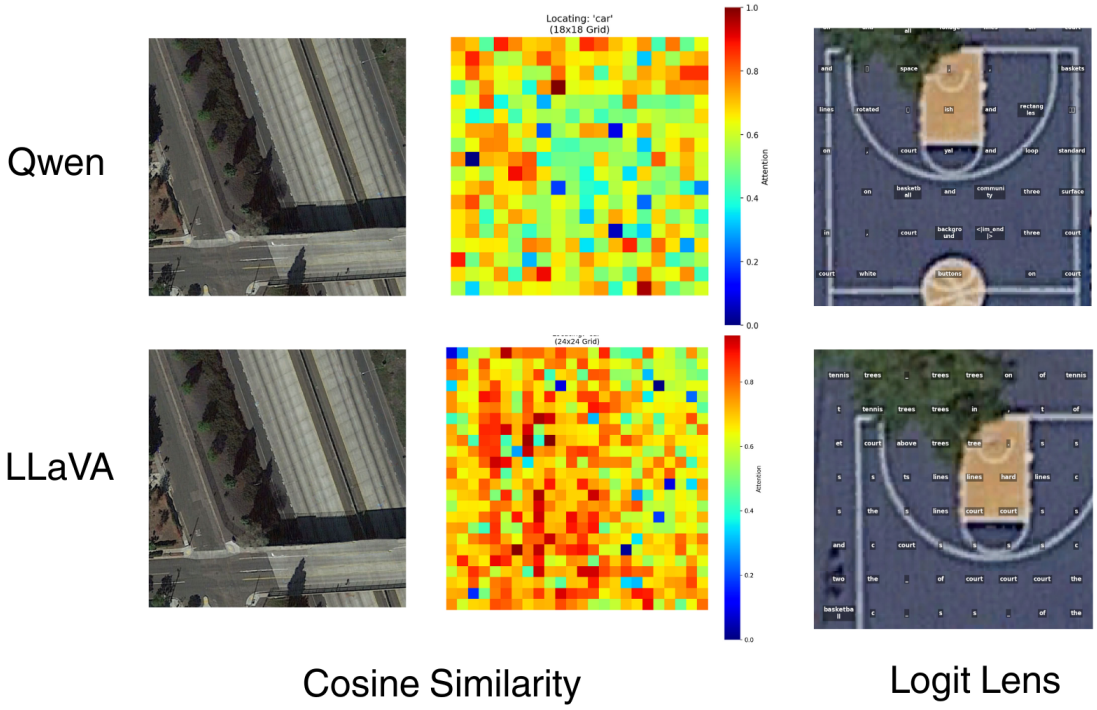


Figure 3: High cosine similarity between the visual tokens and the text embedding of 'car' even though there are no cars on the highway. Supports observation of spurious correlations in visual embeddings leading to hallucinations in LLaVA. Notice the significantly lesser amount of such errors for Qwen. Similarly, LogitLens shows LLaVA predicting tennis court for some tokens for an image of a basketball court.

References

[1] Bai, S.; Chen, K.; Liu, X.; Wang, J.; Ge, W.; Song, S.; Dang, K.; Wang, P.; Wang, S.; Tang, J.; Zhong, H.; Zhu, Y.; Yang, M.; Li, Z.; Wan, J.; Wang, P.; Ding, W.; Fu, Z.; Xu, Y.; Ye, J.; Zhang, X.; Xie, T.; Cheng, Z.; Zhang, H.; Yang, Z.; Xu, H.; and Lin, J. 2025. Qwen2.5-VL Technical Report. arXiv:2502.13923.

[2] Bai, Z.; Wang, P.; Xiao, T.; He, T.; Han, Z.; Zhang, Z.; and Shou, M. Z. 2025. Hallucination of Multimodal Large Language Models: A Survey. arXiv:2404.18930.

[3] Bolya, D.; Huang, P.-Y.; Sun, P.; Cho, J. H.; Madotto, A.; Wei, C.; Ma, T.; Zhi, J.; Rajasegaran, J.; Rasheed, H.; Wang, J.; Monteiro, M.; Xu, H.; Dong, S.; Ravi, N.; Li, D.; Dollár, P.; and Feichtenhofer, C. 2025. Perception Encoder: The best visual embeddings are not at the output of the network. arXiv:2504.13181.

[4] Buglakova, E.; Archit, A.; D'Imprima, E.; Mahamid, J.; Pape, C.; and Kreshuk, A. 2025. Tiling artifacts and trade-offs of feature normalization in the segmentation of large biological images. arXiv:2503.19545.

[5] Carion, N.; Gustafson, L.; Hu, Y.-T.; Debnath, S.; Hu, R.; Suris, D.; Ryali, C.; Alwala, K. V.; Khedr, H.; Huang, A.; Lei, J.; Ma, T.; Guo, B.; Kalla, A.; Marks, M.; Greer, J.; Wang, M.; Sun, P.; Rädle, R.; Afouras, T.; Mavroudi, E.; Xu, K.; Wu, T.-H.; Zhou, Y.; Momeni, L.; Hazra, R.; Ding, S.; Vaze, S.; Porcher, F.; Li, F.; Li, S.; Kamath, A.; Cheng, H. K.; Dollár, P.; Ravi, N.; Saenko, K.; Zhang, P.; and Feichtenhofer, C. 2025. SAM 3: Segment Anything with Concepts. arXiv:2511.16719.

[6] Carion, N.; Massa, F.; Synnaeve, G.; Usunier, N.; Kirillov, A.; and Zagoruyko, S. 2020. End-to-End Object Detection with Transformers. arXiv:2005.12872.

[7] Cheng, Q.; Huang, H.; Xu, Y.; Zhou, Y.; Li, H.; and Wang, Z. 2022. NWPU-Captions Dataset and MLCA-Net for Remote Sensing Image Captioning. *IEEE Transactions on Geoscience and Remote Sensing*, 60: 1–19.

[8] Dehghani, M.; Mustafa, B.; Djolonga, J.; Heek, J.; Minderer, M.; Caron, M.; Steiner, A.; Puigcerver, J.; Geirhos, R.; Alabdulmohsin, I.; Oliver, A.; Padlewski, P.; Gritsenko, A.; Lučić, M.; and Houlsby, N. 2023. Patch n' Pack: NaViT, a Vision Transformer for any Aspect Ratio and Resolution. arXiv:2307.06304.

[9] He, K.; Gkioxari, G.; Dollár, P.; and Girshick, R. 2018. Mask R-CNN. arXiv:1703.06870.

[10] Hosseini, P.; Nawathe, S.; Moayeri, M.; Balasubramanian, S.; and Feizi, S. 2025. Seeing What's Not There: Spurious Correlation in Multimodal LLMs. arXiv:2503.08884.

[11] Hu, Y.; Yuan, J.; Wen, C.; Lu, X.; and Li, X. 2023. RSGPT: A Remote Sensing Vision Language Model and Benchmark. arXiv:2307.15266.

[12] Kim, S.; Lee, A.; Park, J.; Chung, A.; Oh, J.; and Lee, J.-Y. 2024. Towards Efficient Visual-Language Alignment of the Q-Former for Visual Reasoning Tasks. arXiv:2410.09489.

[13] Kirillov, A.; Mintun, E.; Ravi, N.; Mao, H.; Rolland, C.; Gustafson, L.; Xiao, T.; Whitehead, S.; Berg, A. C.; Lo, W.-Y.; Dollár, P.; and Girshick, R. 2023. Segment Anything. arXiv:2304.02643.

[14] Kuckreja, K.; Danish, M. S.; Naseer, M.; Das, A.; Khan, S.; and Khan, F. S. 2023. GeoChat: Grounded Large Vision-Language Model for Remote Sensing. arXiv:2311.15826.

[15] Lai, X.; Tian, Z.; Chen, Y.; Li, Y.; Yuan, Y.; Liu, S.; and Jia, J. 2024. LISA: Reasoning Segmentation via Large Language Model. arXiv:2308.00692.

[16] Li, F.; Zhang, R.; Zhang, H.; Zhang, Y.; Li, B.; Li, W.; Ma, Z.; and Li, C. 2024. LLaVA-NeXT-Interleave: Tackling Multi-image, Video, and 3D in Large Multimodal Models. *arXiv preprint arXiv:2407.07895*.

[17] Li, J.; Li, D.; Xiong, C.; and Hoi, S. 2022. BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation. arXiv:2201.12086.

[18] Li, K.; Wan, G.; Cheng, G.; Meng, L.; and Han, J. 2020. Object detection in optical remote sensing images: A survey and a new benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing*, 159: 296–307.

[19] Li, K. Y.; Goyal, S.; Semedo, J. D.; and Kolter, J. Z. 2025. Inference Optimal VLMs Need Fewer Visual Tokens and More Parameters. arXiv:2411.03312.

[20] Li, X.; Ding, J.; and Elhoseiny, M. 2024. VRS-Bench: A Versatile Vision-Language Benchmark Dataset for Remote Sensing Image Understanding. arXiv:2406.12384.

[21] Liu, H.; Li, C.; Wu, Q.; and Lee, Y. J. 2023. Visual Instruction Tuning. arXiv:2304.08485.

[22] Liu, S.; Ma, Y.; Zhang, X.; Wang, H.; Ji, J.; Sun, X.; and Ji, R. 2024. Rotated Multi-Scale Interaction Network for Referring Remote Sensing Image Segmentation. arXiv:2312.12470.

[23] Liu, S.; Zeng, Z.; Ren, T.; Li, F.; Zhang, H.; Yang, J.; Jiang, Q.; Li, C.; Yang, J.; Su, H.; Zhu, J.; and Zhang, L. 2024. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. arXiv:2303.05499.

[24] Liu, Y.; Ma, Z.; Pu, J.; Qi, Z.; Wu, Y.; Shan, Y.; and Chen, C. W. 2025. UniPixel: Unified Object Referring and Segmentation for Pixel-Level Visual Reasoning. arXiv:2509.18094.

[25] Lobry, S.; Marcos, D.; Murray, J.; and Tuia, D. 2020. RSVQA: Visual Question Answering for Remote Sensing Data. *IEEE Transactions on Geoscience and Remote Sensing*, 58(12): 8555–8566.

[26] Lu, S.; Li, Y.; Chen, Q.-G.; Xu, Z.; Luo, W.; Zhang, K.; and Ye, H.-J. 2024. Ovis: Structural Embedding Alignment for Multimodal Large Language Model. arXiv:2405.20797.

[27] Masry, A.; Rodriguez, J. A.; Zhang, T.; Wang, S.; Wang, C.; Feizi, A.; Suresh, A. K.; Puri, A.; Jian, X.; Noël, P.-A.; Madhusudhan, S. T.; Pedersoli, M.; Liu, B.; Chapados, N.; Bengio, Y.; Hoque, E.; Pal, C.; Laradji, I. H.; Vazquez, D.; Taslakian, P.; Gella, S.; and Rajeswar, S. 2025. AlignVLM: Bridging Vision and Language Latent Spaces for Multimodal Document Understanding. arXiv:2502.01341.

[28] nostalgebraist. 2020. Interpreting GPT: the logit lens. LessWrong blog post. https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

[29] OpenAI; Achiam, J.; Adler, S.; Agarwal, S.; Ahmad, L.; Akkaya, I.; Aleman, F. L.; Almeida, D.; Altenschmidt, J.; Altman, S.; Anadkat, S.; Avila, R.; Babuschkin, I.; Balaji, S.; Balcom, V.; Baltescu, P.; Bao, H.; Bavarian, M.; Belgum, J.; Bello, I.; Berdine, J.; Bernadett-Shapiro, G.; Berner, C.; Bogdonoff, L.; Boiko, O.; Boyd, M.; Brakman, A.-L.; Brockman, G.; Brooks, T.; Brundage, M.; Button, K.; Cai, T.; Campbell, R.; Cann, A.; Carey, B.; Carlson, C.; Carmichael, R.; Chan, B.; Chang, C.; Chantzis, F.; Chen, D.; Chen, S.; Chen, R.; Chen, J.; Chen, M.; Chess, B.; Cho, C.; Chu, C.; Chung, H. W.; Cummings, D.; Currier, J.; Dai, Y.; Decareaux, C.; Degry, T.; Deutsch, N.; Deville, D.; Dhar, A.; Dohan, D.; Dowling, S.; Dunning, S.; Ecoffet, A.; Eleti, A.; Eloundou, T.; Farhi, D.; Fedus, L.; Felix, N.; Fishman, S. P.; Forte, J.; Fulford, I.; Gao, L.; Georges, E.; Gibson, C.; Goel, V.; Gogineni, T.; Goh, G.; Gontijo-Lopes, R.; Gordon, J.; Grafstein, M.; Gray, S.; Greene, R.; Gross, J.; Gu, S. S.; Guo, Y.; Hallacy, C.; Han, J.; Harris, J.; He, Y.; Heaton, M.; Heidecke, J.; Hesse, C.; Hickey, A.; Hickey, W.; Hoeschele, P.; Houghton, B.; Hsu, K.; Hu, S.; Hu, X.; Huizinga, J.; Jain, S.; Jain, S.; Jang, J.; Jiang, A.; Jiang, R.; Jin, H.; Jin, D.; Jomoto, S.; Jonn, B.; Jun, H.; Kaftan, T.; Łukasz Kaiser; Kamali, A.; Kanitscheider, I.; Keskar, N. S.; Khan, T.; Kilpatrick, L.; Kim, J. W.; Kim, C.; Kim, Y.; Kirchner, J. H.; Kiros, J.; Knight, M.; Kokotajlo, D.; Łukasz Kondraciuk; Kondrich, A.; Konstantinidis, A.; Kosic, K.; Krueger, G.; Kuo, V.; Lampe, M.; Lan, I.; Lee, T.; Leike, J.; Leung, J.; Levy, D.; Li, C. M.; Lim, R.; Lin, M.; Lin, S.; Litwin, M.; Lopez, T.; Lowe, R.; Lue, P.; Makanju, A.; Malfacini, K.; Manning, S.; Markov, T.; Markovski, Y.; Martin, B.; Mayer, K.; Mayne, A.; McGrew, B.; McKinney, S. M.; McLeavey, C.; McMillan, P.; McNeil, J.; Medina, D.; Mehta, A.; Menick, J.; Metz, L.; Mishchenko, A.; Mishkin, P.; Monaco, V.; Morikawa, E.; Mossing, D.; Mu, T.; Murati, M.; Murk, O.; Mély, D.; Nair, A.; Nakano, R.; Nayak, R.; Neelakantan, A.; Ngo, R.; Noh, H.; Ouyang, L.; O'Keefe, C.; Pachocki, J.; Paino, A.; Palermo, J.; Pantuliano, A.; Parascandolo, G.; Parish, J.; Parparita, E.; Passos, A.; Pavlov, M.; Peng, A.; Perelman, A.; de Avila Belbute Peres, F.; Petrov, M.; de Oliveira Pinto, H. P.; Michael; Pokorny; Pokrass, M.; Pong, V. H.; Powell, T.; Power, A.; Power, B.; Proehl, E.; Puri, R.; Radford, A.; Rae, J.; Ramesh, A.; Raymond, C.; Real, F.; Rimbach, K.; Ross, C.; Rotsted, B.; Roussez, H.; Ryder, N.; Saltarelli, M.; Sanders, T.; Santurkar, S.; Sastry, G.; Schmidt, H.; Schnurr, D.; Schulman, J.; Selsam, D.; Sheppard, K.; Sherbakov, T.; Shieh, J.; Shoker, S.; Shyam, P.; Sidor, S.; Sigler, E.; Simens, M.; Sitkin, J.; Slama, K.; Sohl, I.; Sokolowsky, B.; Song, Y.; Staudacher, N.; Such, F. P.; Summers, N.; Sutskever, I.; Tang, J.; Tezak, N.; Thompson, M. B.; Tillet, P.; Tootoonchian, A.; Tseng, E.; Tuggle, P.; Turley, N.; Tworek, J.; Uribe, J. F. C.; Vallone, A.; Vijayvergiya, A.; Voss, C.; Wainwright, C.; Wang, J. J.; Wang, A.; Wang, B.; Ward, J.; Wei, J.; Weinmann, C.; Welihinda, A.; Welinder, P.; Weng, J.; Weng, L.; Wiethoff, M.; Willner, D.; Winter, C.; Wolrich, S.; Wong, H.; Workman, L.; Wu, S.; Wu, J.; Wu, M.; Xiao, K.; Xu, T.; Yoo, S.; Yu, K.; Yuan, Q.; Zaremba, W.; Zellers, R.; Zhang, C.; Zhang, M.; Zhao, S.; Zheng, T.; Zhuang, J.; Zhuk, W.; and Zoph, B. 2024. GPT-4 Technical Report. arXiv:2303.08774.

[30] Ou, R.; Hu, Y.; Zhang, F.; Chen, J.; and Liu, Y. 2025. GeoPix: Multi-Modal Large Language Model for Pixel-level Image Understanding in Remote Sensing. arXiv:2501.06828.

[31] Pacini, G.; Bianchi, L.; Ciampi, L.; Messina, N.; Amato, G.; and Falchi, F. 2025. CountingDINO: A Training-free Pipeline for Class-Agnostic Counting us-

ing Unsupervised Backbones. arXiv:2504.16570.

[32] Qin, Y.; Wei, B.; Ge, J.; Kallidromitis, K.; Fu, S.; Darrell, T.; and Wang, X. 2025. Chain-of-Visual-Thought: Teaching VLMs to See and Think Better with Continuous Visual Tokens. arXiv:2511.19418.

[33] Quenum, J.; Hsieh, W.-H.; Wu, T.-H.; Gupta, R.; Darrell, T.; and Chan, D. M. 2025. LISAT: Language-Instructed Segmentation Assistant for Satellite Imagery. arXiv:2505.02829.

[34] Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. arXiv:2103.00020.

[35] Ravi, N.; Gabeur, V.; Hu, Y.-T.; Hu, R.; Ryali, C.; Ma, T.; Khedr, H.; Rädle, R.; Rolland, C.; Gustafson, L.; Mintun, E.; Pan, J.; Alwala, K. V.; Carion, N.; Wu, C.-Y.; Girshick, R.; Dollár, P.; and Feichtenhofer, C. 2024. SAM 2: Segment Anything in Images and Videos. arXiv:2408.00714.

[36] Siméoni, O.; Vo, H. V.; Seitzer, M.; Baldassarre, F.; Oquab, M.; Jose, C.; Khalidov, V.; Szafraniec, M.; Yi, S.; Ramamonjisoa, M.; Massa, F.; Haziza, D.; Wehrstedt, L.; Wang, J.; Darcet, T.; Moutakanni, T.; Sentana, L.; Roberts, C.; Vedaldi, A.; Tolan, J.; Brandt, J.; Couprie, C.; Mairal, J.; Jégou, H.; Labatut, P.; and Bojanowski, P. 2025. DINOv3. arXiv:2508.10104.

[37] Sun, X.; Wang, P.; Yan, Z.; Xu, F.; Wang, R.; Diao, W.; Chen, J.; Li, J.; Feng, Y.; Xu, T.; Weinmann, M.; Hinz, S.; Wang, C.; and Fu, K. 2021. FAIR1M: A Benchmark Dataset for Fine-grained Object Recognition in High-Resolution Remote Sensing Imagery. arXiv:2103.05569.

[38] Team, G.; Kamath, A.; Ferret, J.; Pathak, S.; Vieillard, N.; Merhej, R.; Perrin, S.; Matejovicova, T.; Ramé, A.; Rivière, M.; Rouillard, L.; Mesnard, T.; Cideron, G.; bastien Grill, J.; Ramos, S.; Yvinec, E.; Casbon, M.; Pot, E.; Penchev, I.; Liu, G.; Visin, F.; Kenealy, K.; Beyer, L.; Zhai, X.; Tsitsulin, A.; Busa-Fekete, R.; Feng, A.; Sachdeva, N.; Coleman, B.; Gao, Y.; Mustafa, B.; Barr, I.; Parisotto, E.; Tian, D.; Eyal, M.; Cherry, C.; Peter, J.-T.; Sinopalnikov, D.; Bhupatiraju, S.; Agarwal, R.; Kazemi, M.; Malkin, D.; Kumar, R.; Vilar, D.; Brusilovsky, I.; Luo, J.; Steiner, A.; Friesen, A.; Sharma, A.; Sharma, A.; Gilady, A. M.; Goedeckemeyer, A.; Saade, A.; Feng, A.; Kolesnikov, A.; Bendebury, A.; Abdagic, A.; Vadi, A.; György, A.; Pinto, A. S.; Das, A.; Bapna, A.; Miech, A.; Yang, A.; Paterson, A.; Shenoy, A.; Chakrabarti, A.; Piot, B.; Wu, B.; Shahriari, B.; Petrini, B.; Chen, C.; Lan, C. L.; Choquette-Choo, C. A.; Carey, C.; Brick, C.; Deutsch, D.; Eisenbud, D.; Cattle, D.; Cheng, D.; Paparas, D.; Sreepathihalli, D. S.; Reid, D.; Tran, D.; Zelle, D.; Noland, E.; Huizenga, E.; Kharitonov, E.; Liu, F.; Amirkhanyan, G.; Cameron, G.; Hashemi, H.; Klimczak-Plucińska, H.; Singh, H.; Mehta, H.; Lehri, H. T.; Hazimeh, H.; Ballantyne, I.; Szpektor, I.; Nardini, I.; Pouget-Abadie, J.; Chan, J.; Stanton, J.; Wieting, J.; Lai, J.; Orbay, J.; Fernandez, J.; Newlan, J.; yeong Ji, J.; Singh, J.; Black, K.; Yu, K.; Hui, K.; Vodrahalli, K.; Greff, K.; Qiu, L.; Valentine, M.; Coelho, M.; Ritter, M.; Hoffman, M.; Watson, M.; Chaturvedi, M.; Moynihan, M.; Ma, M.; Babar, N.; Noy, N.; Byrd, N.; Roy, N.; Momchev, N.; Chauhan, N.; Sachdeva, N.; Bunyan, O.; Botarda, P.; Caron, P.; Rubenstein, P. K.; Culliton, P.; Schmid, P.; Sessa, P. G.; Xu, P.; Stanczyk, P.; Tafti, P.; Shivanna, R.; Wu, R.; Pan, R.; Rokni, R.; Willoughby, R.; Vallu, R.; Mullins, R.; Jerome, S.; Smoot, S.; Girgin, S.; Iqbal, S.; Reddy, S.; Sheth, S.; Põder, S.; Bhatnagar, S.; Panyam, S. R.; Eiger, S.; Zhang, S.; Liu, T.; Yacovone, T.; Liechty, T.; Kalra, U.; Evci, U.; Misra, V.; Roseberry, V.; Feinberg, V.; Kolesnikov, V.; Han, W.; Kwon, W.; Chen, X.; Chow, Y.; Zhu, Y.; Wei, Z.; Egyed, Z.; Cotruta, V.; Giang, M.; Kirk, P.; Rao, A.; Black, K.; Babar, N.; Lo, J.; Moreira, E.; Martins, L. G.; Sanseviero, O.; Gonzalez, L.; Gleicher, Z.; Warkentin, T.; Mirrokni, V.; Senter, E.; Collins, E.; Barral, J.; Ghahramani, Z.; Hadsell, R.; Matias, Y.; Sculley, D.; Petrov, S.; Fiedel, N.; Shazeer, N.; Vinyals, O.; Dean, J.; Hassabis, D.; Kavukcuoglu, K.; Farabet, C.; Buchatskaya, E.; Alayrac, J.-B.; Anil, R.; Dmitry; Lepikhin; Borgeaud, S.; Bachem, O.; Joulin, A.; Andreev, A.; Hardin, C.; Dadashi, R.; and Hussenot, L. 2025. Gemma 3 Technical Report. arXiv:2503.19786.

[39] Wang, F.; Chen, M.; Li, Y.; Wang, D.; Wang, H.; Guo, Z.; Wang, Z.; Shan, B.; Lan, L.; Wang, Y.; Wang, H.; Yang, W.; Du, B.; and Zhang, J. 2025. GeoLLaVA-8K: Scaling Remote-Sensing Multimodal Large Language Models to 8K Resolution. arXiv:2505.21375.

[40] Wang, F.; Wang, H.; Chen, M.; Wang, D.; Wang, Y.; Guo, Z.; Ma, Q.; Lan, L.; Yang, W.; Zhang, J.; Liu, Z.; and Sun, M. 2025. XLRS-Bench: Could Your Multimodal LLMs Understand Extremely Large Ultra-High-Resolution Remote Sensing Imagery? arXiv:2503.23771.

[41] Wang, P.; Bai, S.; Tan, S.; Wang, S.; Fan, Z.; Bai, J.; Chen, K.; Liu, X.; Wang, J.; Ge, W.; Fan, Y.; Dang, K.; Du, M.; Ren, X.; Men, R.; Liu, D.; Zhou, C.; Zhou, J.; and Lin, J. 2024. Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution. arXiv:2409.12191.

[42] Xia, G.-S.; Bai, X.; Ding, J.; Zhu, Z.; Belongie, S.; Luo, J.; Datcu, M.; Pelillo, M.; and Zhang, L. 2019. DOTA: A Large-scale Dataset for Object Detection in Aerial Images. arXiv:1711.10398.

[43] Yang, A.; Li, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Gao, C.; Huang, C.; Lv, C.; Zheng, C.; Liu, D.; Zhou, F.; Huang, F.; Hu, F.; Ge, H.; Wei, H.; Lin, H.; Tang, J.; Yang, J.; Tu, J.; Zhang, J.; Yang, J.; Yang, J.; Zhou, J.; Zhou, J.; Lin, J.; Dang, K.; Bao, K.; Yang, K.; Yu, L.; Deng, L.; Li, M.; Xue, M.; Li, M.; Zhang, P.; Wang, P.; Zhu, Q.; Men, R.; Gao, R.; Liu, S.; Luo, S.; Li, T.; Tang, T.; Yin, W.; Ren, X.; Wang, X.; Zhang, X.; Ren, X.; Fan, Y.; Su, Y.; Zhang, Y.;

Zhang, Y.; Wan, Y.; Liu, Y.; Wang, Z.; Cui, Z.; Zhang, Z.; Zhou, Z.; and Qiu, Z. 2025. Qwen3 Technical Report. arXiv:2505.09388.

[44] Yang, J.; Zhang, H.; Li, F.; Zou, X.; Li, C.; and Gao, J. 2023. Set-of-Mark Prompting Unleashes Extraordinary Visual Grounding in GPT-4V. arXiv:2310.11441.

[45] Yao, K.; Xu, N.; Yang, R.; Xu, Y.; Gao, Z.; Kitrungrotsakul, T.; Ren, Y.; Zhang, P.; Wang, J.; Wei, N.; and Li, C. 2025. Falcon: A Remote Sensing Vision-Language Foundation Model (Technical Report). arXiv:2503.11070.

[46] Zamir, S. W.; Arora, A.; Gupta, A.; Khan, S.; Sun, G.; Khan, F. S.; Zhu, F.; Shao, L.; Xia, G.-S.; and Bai, X. 2019. iSAID: A Large-scale Dataset for Instance Segmentation in Aerial Images. arXiv:1905.12886.

[47] Zamir, S. W.; Arora, A.; Gupta, A.; Khan, S.; Sun, G.; Khan, F. S.; Zhu, F.; Shao, L.; Xia, G.-S.; and Bai, X. 2019. iSAID: A Large-scale Dataset for Instance Segmentation in Aerial Images. arXiv:1905.12886.

[48] Zhou, Y.; Lan, M.; Li, X.; Feng, L.; Ke, Y.; Jiang, X.; Li, Q.; Yang, X.; and Zhang, W. 2025. GeoGround: A Unified Large Vision-Language Model for Remote Sensing Visual Grounding. arXiv:2411.11904.

[49] Zi, X.; Xiao, J.; Shi, Y.; Tao, X.; Li, J.; Braytee, A.; and Prasad, M. 2025. RSVLM-QA: A Benchmark Dataset for Remote Sensing Vision Language Model-based Question Answering. arXiv:2508.07918.